Handling Data 2 — Frequency tables · Discrete and continuous data

Calculator allowed. Squared paper for the frequency polygons. Marks in brackets. ★ = harder question.

1 Dani records the number of goals (x) she scores per game over a season of 20 games. [5]

x	0	1	2	3	4	5	6
Frequency	1	4	6	5	2	1	1

Last season her mean was 2.1 goals per game. Show that she has improved this season, and write down this season's mode and median.

2 The times (t minutes) taken by 20 students to finish a puzzle were: [10]

3, 16, 5, 19, 6, 7, 10, 8, 13, 12, 9, 7, 10, 11, 12, 16, 15, 13, 8, 5

a Make a frequency table with classes $0 \le t < 5$, $5 \le t < 10$, $10 \le t < 15$, $15 \le t \le 20$.

b Draw a frequency polygon.

c Use the table to estimate the mean time, and state the modal class.

d Calculate the exact mean from the raw data and explain why it differs from your estimate.

3 A survey of 100 teenagers asked how many hours (t) they slept on New Year's Day. [10]

Hours t	$0 \le t < 4$	$4 \le t < 6$	$6 \le t < 8$	$8 \le t \le 12$
Frequency	p	24	$3p$	16

a Find p and complete the table.

b Draw a frequency polygon.

c Estimate the mean and state the modal class.

d Estimate the percentage of teenagers who slept at least 7 hours.

4 The mean of the numbers 4, 7, x, 9 and 11 is 8. Find x, and then find the median. [3]

Total: 28 marks

Check: mean from a table $= \frac{\sum fx}{\sum f}$ (divide by the total frequency, not the number of rows) · grouped data → mid-points, and the mean is an *estimate* · modal class = highest frequency · frequency polygon: points at mid-points, joined by straight lines.